

Phase 6 - Categorising unseen samples

Single images are NOT reliably categorisable: LOO which-batch accuracy B1 43%/B2 14%/B3 53% with a 65% false-alarm rate on held-out Batch_3 (artefact-safe set: 29%/0%/65%). Groups of ≥ 5 images reach ~ 97 -100% for Batch_1; Batch_2 needs ~ 7 because it genuinely overlaps Batch_3. categorise.py therefore treats a folder as one sample (median features) and sizes the OUT threshold to the group count.

Assumption (applies throughout): the grey bulk is assumed graphite, bright particles are assumed silicon-based, black is pore/crack. This comes from image appearance and is unconfirmed.

LOO predictions (all kept features)

image_id	batch	dist	threshold	call_inout	pred_batch	conf	drivers	feats
4ih2ggld	Batch_1	12.98	3.53	OUT	Batch_1	0.9189	silicon_frac:14.05 corr_len_gr_um:4.04 etd	si_d10_um corr_len_gr_um silicon_frac etd
5n1q8atc	Batch_1	14.86	4.876	OUT	Batch_1	0.8425	silicon_frac:16.07 etd_roughness:2.47 bse	si_d10_um etd_roughness bse_bulk_texture g
f1vzngrs	Batch_1	3.969	3.7	OUT	Batch_1	0.5076	silicon_frac:2.81 si_d10_um:1.36 corr_len	silicon_frac si_d10_um corr_len_gr_um etd
ffwubibz	Batch_1	1.372	3.3	IN	Batch_3	0.7421	silicon_frac:1.63 si_d10_um:1.14 si_clarke	silicon_frac si_d10_um corr_len_gr_um etd
fzrt2k6r	Batch_1	2.811	3.53	IN	Batch_2	0.5463	etd_roughness:1.03 silicon_frac:0.95 bse_b	silicon_frac si_d10_um corr_len_gr_um etd
iv6g2oq0	Batch_1	1.956	3.53	IN	Batch_2	0.5652	bse_bulk_texture:1.03 pores_per_mm2:0.99 e	silicon_frac si_d10_um corr_len_gr_um etd
uhdslk0o	Batch_1	1.912	3.53	IN	Batch_2	0.5963	corr_len_gr_um:1.35 bse_bulk_texture:0.82	silicon_frac si_d10_um etd_roughness corr
3806gxp0	Batch_2	1.336	3.53	IN	Batch_3	0.7517	corr_len_gr_um:0.67 etd_roughness:0.48 por	si_d10_um silicon_frac corr_len_gr_um etd
avn74qx1	Batch_2	2.027	3.53	IN	Batch_3	0.8245	silicon_frac:2.15 si_d10_um:1.51 pores_per	si_d10_um silicon_frac corr_len_gr_um etd
b3esycq1	Batch_2	3.782	3.53	OUT	Batch_1	0.8742	silicon_frac:3.48 corr_len_gr_um:2.02 etd	si_d10_um silicon_frac corr_len_gr_um etd
epqdaau9	Batch_2	7.183	4.533	OUT	Batch_2	0.4932	pores_per_mm2:2.38 gr_st_coherence:1.39 et	silicon_frac corr_len_gr_um etd_roughness
i9jiqjwl	Batch_2	2.528	3.53	IN	Batch_3	0.7325	silicon_frac:1.70 etd_roughness:0.81 bse_b	si_d10_um silicon_frac corr_len_gr_um etd
r17byphk	Batch_2	1.496	3.53	IN	Batch_3	0.5946	silicon_frac:0.98 bse_bulk_texture:0.75 et	si_d10_um silicon_frac corr_len_gr_um etd
rxax5ozo	Batch_2	2.997	3.408	IN	Batch_3	0.5984	pores_per_mm2:1.22 si_d10_um:0.80 etd_roug	silicon_frac si_d10_um corr_len_gr_um etd
0grcilhi	Batch_3	2.829	4.515	IN	Batch_2	0.5954	gr_st_coherence:2.07 corr_len_gr_um:1.35 s	silicon_frac si_d10_um corr_len_gr_um etd
71vgq3fw	Batch_3	2.237	3.263	IN	Batch_3	0.8446	bse_bulk_texture:2.57 silicon_frac:1.71 po	silicon_frac si_d10_um etd_roughness corr
9luzk4jm	Batch_3	3.163	3.39	IN	Batch_3	0.9087	bse_bulk_texture:2.36 si_d10_um:2.19 pores	silicon_frac si_d10_um corr_len_gr_um etd
cfe5vt7s	Batch_3	3.898	3.855	OUT	Batch_2	0.7112	silicon_frac:1.39 si_d10_um:1.17 etd_rough	silicon_frac si_d10_um etd_roughness corr
hawkfj64	Batch_3	5.975	3.47	OUT	Batch_2	0.5735	gr_st_coherence:2.68 corr_len_gr_um:2.02 s	silicon_frac si_d10_um corr_len_gr_um etd
hzumfsms	Batch_3	3.607	3.371	OUT	Batch_3	0.8462	si_d10_um:2.36 bse_bulk_texture:2.17 corr	silicon_frac si_d10_um corr_len_gr_um etd
kbdh4tri	Batch_3	3.424	3.262	OUT	Batch_3	0.619	silicon_frac:4.06 corr_len_gr_um:2.70 bse	silicon_frac si_d10_um corr_len_gr_um pore
mgxahqnk	Batch_3	5.873	5.174	OUT	Batch_2	0.6077	gr_st_coherence:2.48 silicon_frac:1.36 cor	silicon_frac si_d10_um corr_len_gr_um etd
pl8uabbv	Batch_3	2.05	3.652	IN	Batch_3	0.6227	etd_roughness:1.00 bse_bulk_texture:0.55 c	silicon_frac si_d10_um etd_roughness corr
ptg8lmto	Batch_3	5.53	3.879	OUT	Batch_2	0.5473	corr_len_gr_um:1.35 etd_roughness:1.22 sil	silicon_frac si_d10_um etd_roughness corr
tuy3zymq	Batch_3	4.84	3.371	OUT	Batch_1	0.6237	si_solidity:5.90 silicon_frac:4.63 corr_le	silicon_frac si_d10_um corr_len_gr_um etd
ufdvpb81	Batch_3	3.563	3.4	OUT	Batch_3	0.9072	bse_bulk_texture:2.56 si_d10_um:1.21 pores	silicon_frac si_d10_um corr_len_gr_um etd
utfgcjfa	Batch_3	2.651	3.743	IN	Batch_3	0.8043	si_d10_um:1.20 etd_roughness:1.03 pores_pe	silicon_frac si_d10_um etd_roughness corr

image_id	batch	dist	threshold	call_inout	pred_batch	conf	drivers	feats
vc2whyaq	Batch_3	5.08	3.829	OUT	Batch_3	0.8008	si_d10_um:4.06 pores_per_mm2:1.71 etd_roug	si_d10_um silicon_frac etd_roughness corr_
x77cy643	Batch_3	4.424	3.798	OUT	Batch_3	0.7797	si_d10_um:2.72 etd_roughness:1.30 pores_pe	silicon_frac si_d10_um etd_roughness corr_
x7u69zsw	Batch_3	4.091	3.25	OUT	Batch_1	0.6782	silicon_frac:5.16 bse_bulk_texture:2.41 po	silicon_frac si_d10_um corr_len_gr_um etd_
xgj4xftb	Batch_3	3.591	3.697	IN	Batch_2	0.4354	corr_len_gr_um:1.35 pores_per_mm2:1.30 etd	silicon_frac si_d10_um corr_len_gr_um etd_

LOO predictions (artefact-safe features)

image_id	batch	dist	threshold	call_inout	pred_batch	conf	drivers	feats
4ih2ggld	Batch_1	11.17	3.297	OUT	Batch_1	0.9084	silicon_frac:14.05 corr_len_gr_um:4.04 por	si_d10_um corr_len_gr_um silicon_frac pore
5n1q8atc	Batch_1	15.65	3.509	OUT	Batch_1	0.8572	silicon_frac:16.07 corr_len_gr_um:8.09 si_	si_d10_um gr_st_coherence silicon_frac inl
f1vzngrs	Batch_1	5.384	3.591	OUT	Batch_3	0.5332	silicon_frac:2.81 gr_st_coherence:1.54 si_	silicon_frac si_d10_um corr_len_gr_um si_c
ffwubibz	Batch_1	2.727	3.174	IN	Batch_3	0.6307	graphite_frac:2.34 silicon_frac:1.63 si_d1	silicon_frac si_d10_um corr_len_gr_um pore
fzrt2k6r	Batch_1	1.191	3.649	IN	Batch_3	0.6764	silicon_frac:0.95 corr_len_gr_um:0.67 gr_s	silicon_frac si_d10_um corr_len_gr_um pore
iv6g2oq0	Batch_1	2.341	3.649	IN	Batch_3	0.5407	gr_st_coherence:1.18 pores_per_mm2:0.99 si	silicon_frac si_d10_um corr_len_gr_um pore
uhdslk0o	Batch_1	2.242	3.649	IN	Batch_3	0.5599	corr_len_gr_um:1.35 si_clarkevans_R:0.94 s	silicon_frac si_d10_um corr_len_gr_um pore
3806gxp0	Batch_2	2.897	3.649	IN	Batch_3	0.6606	gr_st_coherence:2.06 corr_len_gr_um:0.67 s	si_d10_um silicon_frac corr_len_gr_um pore
avn74qx1	Batch_2	2.817	3.226	IN	Batch_3	0.8404	silicon_frac:2.15 si_d10_um:1.51 pores_per	si_d10_um silicon_frac corr_len_gr_um pore
b3esycq1	Batch_2	3.809	3.649	OUT	Batch_1	0.8605	silicon_frac:3.48 corr_len_gr_um:2.02 gr_s	si_d10_um silicon_frac corr_len_gr_um pore
epqdaau9	Batch_2	5.053	3.077	OUT	Batch_3	0.521	pores_per_mm2:2.38 gr_st_coherence:1.39 si	silicon_frac corr_len_gr_um pores_per_mm2
i9jiqjwl	Batch_2	3.157	3.649	IN	Batch_3	0.7513	silicon_frac:1.70 gr_st_coherence:1.63 si_	si_d10_um silicon_frac corr_len_gr_um pore
r17byphk	Batch_2	1.269	3.649	IN	Batch_3	0.7233	silicon_frac:0.98 gr_st_coherence:0.78 si_	si_d10_um silicon_frac corr_len_gr_um pore
rxax5ozo	Batch_2	2.756	3.649	IN	Batch_3	0.6206	pores_per_mm2:1.22 si_d10_um:0.80 corr_len	silicon_frac si_d10_um corr_len_gr_um gr_s
0grcilhi	Batch_3	3.325	3.587	IN	Batch_2	0.627	gr_st_coherence:2.07 corr_len_gr_um:1.35 p	silicon_frac si_d10_um corr_len_gr_um gr_s
71vgq3fw	Batch_3	1.553	3.544	IN	Batch_3	0.7846	silicon_frac:1.71 pores_per_mm2:1.08 gr_st	silicon_frac si_d10_um corr_len_gr_um pore
9luzk4jm	Batch_3	6.277	3.447	OUT	Batch_3	0.6294	ring_porosity_250nm:3.93 si_d10_um:2.19 po	silicon_frac si_d10_um corr_len_gr_um pore
cfe5vt7s	Batch_3	2.009	3.678	IN	Batch_2	0.5692	silicon_frac:1.39 si_d10_um:1.17 gr_st_coh	silicon_frac si_d10_um corr_len_gr_um pore
hawkfj64	Batch_3	7.26	3.393	OUT	Batch_2	0.5819	gr_st_coherence:2.68 corr_len_gr_um:2.02 s	silicon_frac si_d10_um corr_len_gr_um gr_s
hzumfsms	Batch_3	2.781	3.563	IN	Batch_3	0.7819	si_d10_um:2.36 corr_len_gr_um:2.02 si_clar	silicon_frac si_d10_um corr_len_gr_um pore
kbdh4tri	Batch_3	3.377	3.537	IN	Batch_3	0.4198	silicon_frac:4.06 corr_len_gr_um:2.70 pore	silicon_frac si_d10_um corr_len_gr_um pore
mgxahqnk	Batch_3	3.703	3.628	OUT	Batch_2	0.619	gr_st_coherence:2.48 silicon_frac:1.36 cor	silicon_frac si_d10_um corr_len_gr_um gr_s
pl8uabbv	Batch_3	3.08	3.553	IN	Batch_3	0.6413	gr_st_coherence:1.32 si_clarkevans_R:1.31	silicon_frac si_d10_um corr_len_gr_um pore
ptg8lmt0	Batch_3	3.958	3.616	OUT	Batch_3	0.6761	corr_len_gr_um:1.35 gr_st_coherence:0.76 s	silicon_frac si_d10_um corr_len_gr_um pore
tuy3zymq	Batch_3	4.86	3.375	OUT	Batch_1	0.6971	si_solidity:5.90 silicon_frac:4.63 corr_le	silicon_frac si_d10_um corr_len_gr_um si_s
ufdvpb81	Batch_3	2.731	3.235	IN	Batch_3	0.8343	si_d10_um:1.21 inlens_bulk_texture:1.17 po	silicon_frac si_d10_um corr_len_gr_um pore
utfgcjfa	Batch_3	3.225	3.668	IN	Batch_3	0.8335	si_clarkevans_R:1.31 si_d10_um:1.20 pores_	silicon_frac si_d10_um corr_len_gr_um pore
vc2whyaq	Batch_3	5.63	3.378	OUT	Batch_3	0.8297	si_d10_um:4.06 pores_per_mm2:1.71 si_clark	si_d10_um silicon_frac corr_len_gr_um pore
x77cy643	Batch_3	4.344	3.553	OUT	Batch_3	0.8133	si_d10_um:2.72 inlens_edge_density:0.98 po	silicon_frac si_d10_um corr_len_gr_um pore

image_id	batch	dist	threshold	call_inout	pred_batch	conf	drivers	feats
x7u69zsw	Batch_3	3.494	3.585	IN	Batch_1	0.9149	silicon_frac:5.16 si_clarkevans_R:1.20 gr_	silicon_frac si_d10_um corr_len_gr_um pore
xgj4xftb	Batch_3	3.149	3.621	IN	Batch_3	0.596	corr_len_gr_um:1.35 pores_per_mm2:1.30 si_	silicon_frac si_d10_um corr_len_gr_um pore

Accuracy vs group size (which-batch)

batch	n_images	group_accuracy
Batch_1	1	0.42
Batch_1	2	0.555
Batch_1	3	0.775
Batch_1	4	0.83
Batch_1	5	0.97
Batch_1	6	1
Batch_1	7	1
Batch_2	1	0.145
Batch_2	2	0.255
Batch_2	3	0.47
Batch_2	4	0.59
Batch_2	5	0.695
Batch_2	6	0.815
Batch_2	7	1

